

FARNBOROUGH, United Kingdom

WMO: 037680

Lat: 51.2800N

Lon: 0.7725W

Elev: 65

StdP: 100.55

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.8	(c) -3.2	(d) -6.2	(e) 2.3	(f) -2.0	(g) -4.9	(h) 2.5	(i) -0.9	(j) 12.4	(k) 10.4	(l) 11.3	(m) 10.3	(n) 0.9	(o) 230	(p) 0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	27.4	18.9	25.3	18.1	23.5	17.2	20.3	25.0	19.1	23.6	18.1	22.0	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.6	22.1	17.5	12.6	20.8	16.6	11.9	19.8	58.5	24.8	54.5	23.6	51.3	22.1	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	8.9	8.0	-8.0	30.4	1.7	2.2	-9.2	32.0	-10.1	33.3	-11.1	34.5	-12.3	36.1
			-8.2	22.7	2.0	2.2	-9.6	24.3	-10.7	25.6	-11.8	26.8	-13.3	28.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	4.7	5.0	6.6	9.1	12.2	15.3	17.5	17.0	14.3	11.2	7.6	5.2
	DBStd	5.49	3.53	3.36	3.09	3.06	2.96	2.54	2.49	2.50	2.50	3.08	3.41	3.85
	HDD10.0	740	166	142	109	51	12	0	0	0	1	23	86	152
	HDD18.3	2896	422	373	362	276	192	96	47	56	123	221	323	406
	CDD10.0	929	1	2	6	24	79	159	233	218	129	61	13	4
	CDD18.3	44	0	0	0	0	1	5	21	16	2	1	0	0
	CDH23.3	441	0	0	0	2	13	68	199	139	17	2	0	0
	CDH26.7	87	0	0	0	0	0	11	43	33	1	0	0	0
Wind	WSAvg	3.6	4.1	4.1	4.0	3.6	3.7	3.5	3.3	3.2	3.2	3.5	3.5	3.8
Precipitation	PrecAvg	708	75	53	43	53	48	51	49	58	57	74	80	69
	PrecMax	949	184	128	113	141	105	128	122	114	116	154	159	130
	PrecMin	524	11	4	7	3	4	13	12	9	10	29	30	13
	PrecStd	114	37	33	23	34	25	31	24	33	29	34	39	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.0	17.7	22.6	24.9	28.1	30.0	30.0	25.6	20.7	16.0	13.9
		MCWB	11.7	10.7	11.5	14.3	16.3	18.7	19.9	20.1	18.6	16.6	14.4	12.3
	2%	DB	12.0	12.2	14.8	19.1	22.7	25.2	27.7	27.0	22.9	18.1	14.6	13.0
		MCWB	10.6	10.3	10.4	12.8	15.8	17.7	19.0	19.1	17.6	15.1	13.3	11.8
	5%	DB	11.0	11.1	13.1	16.7	20.5	23.0	25.6	24.4	20.8	16.9	13.7	12.1
		MCWB	9.7	9.4	9.9	11.5	14.7	16.7	18.4	18.1	16.3	14.5	12.5	11.0
	10%	DB	10.0	10.1	11.8	14.7	18.0	21.2	23.4	22.3	19.3	15.9	12.8	11.1
		MCWB	8.8	8.5	9.2	10.5	13.3	15.7	17.4	17.1	15.5	14.0	11.5	9.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	11.9	12.8	15.6	18.3	20.7	22.1	22.3	20.1	17.2	14.8	12.8
		MCDB	12.7	13.1	15.5	20.3	22.4	25.8	26.1	26.6	23.3	19.6	15.5	13.3
	2%	WB	11.0	10.7	11.5	13.5	16.5	18.7	20.2	20.3	18.5	16.0	13.7	12.0
		MCDB	11.9	11.7	13.5	18.1	21.0	23.7	25.4	24.6	21.5	17.5	14.4	12.8
	5%	WB	9.9	9.7	10.6	12.2	15.2	17.3	19.0	19.0	17.2	15.1	12.7	11.2
		MCDB	10.8	10.8	12.4	15.3	19.2	21.6	24.1	22.7	19.7	16.4	13.5	12.1
	10%	WB	9.0	8.7	9.7	11.2	14.0	16.3	18.0	17.9	16.2	14.3	11.6	10.0
		MCDB	9.9	10.0	11.5	14.0	17.4	20.2	22.4	21.3	18.4	15.6	12.6	11.0
Mean Daily Temperature Range	5% DB	MDBR	5.9	6.6	8.0	9.3	9.4	9.5	9.8	9.5	9.4	7.8	6.6	5.8
		MCDBR	5.8	7.2	10.2	12.8	13.3	13.5	13.9	13.3	11.7	8.1	6.0	5.2
		MCWBR	5.7	5.7	6.9	7.5	7.3	7.4	7.1	6.8	7.2	5.6	5.3	5.1
	5% WB	MCDBR	6.1	6.4	8.5	11.0	11.7	11.6	11.9	11.0	10.1	7.0	6.1	5.3
		MCWBR	6.1	5.6	6.3	7.1	7.2	7.2	7.0	6.9	7.2	5.5	5.7	5.3
Clear-Sky Solar Irradiance	taub	0.329	0.345	0.382	0.403	0.406	0.405	0.399	0.391	0.375	0.358	0.332	0.319	
	taud	2.366	2.373	2.266	2.227	2.259	2.289	2.312	2.354	2.393	2.419	2.410	2.354	
	Ebn at Noon	679	775	807	834	848	851	850	838	810	749	672	625	
	Edh at Noon	64	82	109	126	128	126	121	110	95	77	61	55	
All-Sky Solar Radiation	RadAvg	0.82	1.49	2.61	4.06	4.91	5.43	5.15	4.29	3.22	1.85	0.98	0.65	
	RadStd	0.07	0.19	0.35	0.45	0.31	0.54	0.48	0.34	0.26	0.16	0.11	0.08	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (82 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page